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How to present food images on packaging? The influence of aesthetics design on consumers' perceived healthiness

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ABSTRACT

Product images on packaging are widespread and essential marketing tools in retailing. Consumers increasingly rely on packaging designs to infer healthiness of food. However, it remains unclear whether the presentation style of food images on packaging influences the perceived healthiness. This study delves into the impact of classical aesthetics design in presenting food images on healthiness judgments, as well as its mechanism and boundary, through five experiments. The results indicate that high classical aesthetic presentation of food images enhances consumers' healthiness perception and purchase intention, with perceived naturalness playing a mediating role. Moreover, including organic label on packaging moderates this aesthetics effect. Specifically, when an organic label is present, perceived naturalness fails to mediate the effect of aesthetics on perceived healthiness. These findings enrich the literature about effects of product images and expand aesthetics research in packaging field, while also offer insights for optimizing packaging marketing strategies and promoting consumers' healthy decision-making.

1. Introduction

Nowadays, healthy diet has become an important concern for most people. Since the nutrients and energy contained in food are intangible, it is challenging for consumers to accurately judge the food healthiness. Therefore, consumers usually rely on various external cues to make healthiness judgments (Chan and Zhang, 2022; Grandi et al., 2021). Product packaging, being the primary attention-grabber for consumers, plays a crucial role in shaping their initial impressions of products, conveying information about product attributes and finally influencing purchases decision (Baek et al., 2023; Chu et al., 2021; Krishna et al., 2017). Previous literature has revealed that some visual designs of food packaging (e.g., color, shape, material) significantly influence consumers' healthiness perception and subsequent consumption decision (Chan and Zhang, 2022; Motoki and Togawa, 2022). For example, consumers perceive foods in lighter colored or less color-saturated packaging (Mai et al., 2016; Mead and Richerson, 2018), slimmer packaging (Koo and Suk, 2016; Yarar et al., 2019) and matte-surface packaging (Ye et al., 2020a) as healthier. In addition to these holistic design elements, some specific details on the packaging, such as the product images, can also influence consumers. As commonly acknowledged, the strategic placement of product images on packaging fronts serves as the potent means of communication to captivate attention, effectively inform essential product details, and shape expectations regarding their attributes (Huang et al., 2022; Spence et al., 2022), ultimately influencing attitudes towards the brand (Underwood and Klein, 2002). Nevertheless, the relationship between food image designs exhibited on packaging and healthiness perception has not been extensively investigated.

Some studies have primarily examined the impacts of images' content characteristics (Mizutani et al., 2010; Yu et al., 2022), amount (Capelli and Thomas, 2021; Madzharov and Block, 2010), size (Huang et al., 2022; Petit et al., 2018) and location (Deng and Kahn, 2009; Togawa et al., 2019) on consumers' attributes perception. The majority of these studies indicate that food images influence perceived tastiness and freshness, while only a handful of studies investigate the image's location effect on healthy behaviors (Deng and Kahn, 2009; Togawa et al., 2019). For example, positioning food images at the bottom of packaging could facilitate healthier eating choices as consumers perceived flavor heaviness (Togawa et al., 2019). Since the image is an

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essential element on the packaging, it is important to uncover additional image designs that can affect the healthiness perception. In practice, we observe diverse presentations of food images on packaging; some are haphazardly arranged while others are neatly organized. Furthermore, the presentation of food holds practical significance within restaurant and advertising contexts. Prior studies have discovered that the neat (vs. non-neat) presentation (Quach et al., 2023; Zellner et al., 2011), vertical (vs. horizontal) presentation (Rowley and Spence, 2018; Szocs and Lefebvre, 2017) and traditional (vs. contemporary) presentation (Zellner et al., 2014) can affect consumers' perception and evaluation of the foods. However, the research exploring the potential effects of presentation styles of food images displayed on packaging, particularly their effects on healthiness perception, remains limited.

Our study mainly focuses on the aesthetics design of the food image presentation. The significance of aesthetics in the realm of product design has garnered widespread recognition (Luchs et al., 2016), and research on aesthetics in marketing has received substantial attention (Burnap et al., 2023; Lee, 2022; Patrick, 2016). Aesthetics has been applied in the packaging research, such as some studies found that high aesthetic food packaging could improve the product's value and stimulate subsequent purchase intentions (Hubert et al., 2013; Reimann et al., 2010). Besides, prior research indicated that food aesthetics can impact consumers' perceptions of food attributes (e.g., healthiness, tastiness), thereby influencing their consumption intentions (Hagen, 2021; Yu and Liu, 2023). However, whether varying aesthetic presentations of food images on packaging affect healthiness judgments remains unidentified.

Therefore, our study specifically focuses on classical aesthetics, which has been extensively researched for its impacts on consumers' perceptions and behaviors across diverse contexts, encompassing web design (Ye et al., 2020b), app design (Bhandari et al., 2019) and food design (Hagen, 2021). In particular, we conducted five experiments to investigate how the classical aesthetic presentation of food images on packaging impacts consumers' perceived healthiness. Furthermore, we examined the mediating role of perceived naturalness and the moderating role of exhibiting organic label on the packaging. Our study contributes to the existing literature through enriching the research on the effects of product images and extending the application of aesthetics theory in packaging research, while also identifying a novel factor moderating the aesthetics effect on perceived healthiness. Moreover, our study enhances the understanding of the cross-modal influence by demonstrating that visual aesthetic cues can affect the healthiness perception. Finally, our study highlights the practical insights for firms to enhance packaging designs and enables consumers to accurately assess healthiness.

2. Theoretical background and hypotheses development

2.1. Effects of food packaging elements on perceived healthiness

Sensory elements in retail environments considerably influence consumers' perceptions, judgments and purchase behaviors (Biswas, 2019). Retailers always leverage sensory interventions to encourage consumers to choose healthier food (Grandi et al., 2021; Luomala et al., 2023; Phillips et al., 2024), such as playing higher-pitched music in grocery stores (Huang and Labroo, 2020), emitting indulgent food-related scent to compensate for the desire for gustatory pleasure (Biswas and Szocs, 2019), and more directly enhancing the visual aesthetics of packaging (Ton et al., 2024). Packaging design refers to a holistic design that integrates different elements to produce sensory effect (Orth and Malkewitz, 2008). Packaging elements are basic components that make up a package, typically divided into visual elements (e.g., color, shape, image) and textual elements (e.g., names, claims, labels), which significantly influence consumers' psychological responses and behaviors (Festila and Chrysochou, 2018). Revolving around the food context, research has shown that multiple visual elements of food packaging can influence consumers' healthiness perceptions (Chan and Zhang, 2022; Goukens and Klesse, 2022; Motoki and Togawa, 2022). Specifically, consumers tend to associate lighter-colored and less color-saturated packaging with healthier foods (Mai et al., 2016; Mead and Richerson, 2018). However, there was other study suggesting that full-color packaging can enhance perceived healthiness (Kunz et al., 2019). In terms of shape, several studies have indicated that taller and slimmer packaging (Koo and Suk, 2016; Yarar et al., 2019) or a dominant shape of packaging (Qian et al., 2023) can enhance healthiness perception. In addition, the packaging material also affects healthiness perception, as research suggests that consumers perceive food in matte packaging as healthier compared to that in glossy packaging (Ye et al., 2020a).

What is more, product images featured on packaging have garnered attention from researchers as a crucial medium for helping consumers comprehend product information (Gil-Pérez et al., 2020; Spence et al., 2022). Prior research has explored the impacts of content characteristics, amount, size and location of product images on consumers' perceptions and behaviors. For instance, incorporating images of fresh oranges on juice packaging enhances consumers' perceptions of freshness (Mizutani et al., 2010) and the images of orange with a juicier appearance significantly enhance perceived juiciness as well as expectations regarding tastiness and quality (Di Cicco et al., 2021). Besides, the design showing the implied motion of food images on the packaging can attract more visual attention, serially increase the freshness, tastiness, product attractiveness and finally purchase intention (Yu et al., 2022). As regard to the amount and size of images, a study found the greater number of food units displayed on the packaging made consumers perceive a greater quantity of food inside and consequently consume more (Madzharov and Block, 2010). Exposure to larger-sized food images on packaging also resulted in increased consumption among individuals (Neyens et al., 2015). Furthermore, the illusion created by larger portions depicted on packaging can enhance consumers' appetizing perception and purchase intention (Petit et al., 2018). Moreover, several studies have also found a location effect of product images, for example, when placed on the lower or right side of packaging, food images elicits heaviness perception compared to those placed on the top or left side (Deng and Kahn, 2009). Building upon this association, subsequent research has revealed the cross-modal influence of visual position of food images on tastiness perception, demonstrating that placing food images at the bottom of packaging enhances consumers' expectations regarding flavor heaviness and encourages them to choose healthier food (Togawa et al., 2019). Therefore, it is preferable to place low-calorie or healthy foods on the left or top side, while high-calorie or unhealthy foods are more suitable for placement on the right or bottom side (Manippa et al., 2022; Romero and Biswas, 2016).

In summary, existing research primarily demonstrates that product images displayed on packaging can influence attributes perception and consumption behaviors. Most of them focus on the effect of food images on the appetite attributes, such as perceived tastiness, while only a few studies have explored the relationship between image position and healthy behaviors. However, there is a lack of direct investigation into whether and how other design elements, such as image presentation, impact consumers' perceived healthiness.

2.2. Effect of aesthetic presentation of food images on perceived healthiness

Previous literature suggests that consumers' perceptions of food attributes can be affected by its presentation, including factors such as neatness (Quach et al., 2023; Zellner et al., 2011), vertical arrangement (Rowley and Spence, 2018; Szocs and Lefebvre, 2017) and modern style (Zellner et al., 2014). For instance, a neat presentation can increase the perceived quality of the food and consequent purchase likelihood (Quach et al., 2023). Besides, when food is stacked on the plate, compared to being spread across it, consumers tend to perceive smaller

portions and consequently consume more (Szocs and Lefebvre, 2017). However, it remains unclear whether the way in which food images displayed on the front of packaging can influence the healthiness perception. Our study focuses on the effect of aesthetic presentation of food images.

Aesthetics is an important element in packaging design (Luchs et al., 2016), while prior studies have largely concentrated on consumers' processing of the holistic aesthetics of packaging and its impact on consumer behaviors (Hubert et al., 2013; Reimann et al., 2010). For instance, Reimann et al. (2010) conducted experiments with fMRI method to reveal the effects of packaging aesthetics on consumers' product selection and corresponding neural mechanisms. Their findings indicated that consumers spent more time choosing aesthetic packaging than standardized ones. Moreover, they revealed that the reward system was activated during the processing of aesthetic packaging and exhibited a positive relationship with affective involvement. Additionally, previous research has verified the neural processing of aesthetic packaging is related to consumers' purchasing behaviors (Hubert et al., 2013; Reimann et al., 2010).

Consumers rely on the food aesthetics to judge the attributes, and highly appealing food is associated with lower perceived risks (Castagna et al., 2021) and higher perceived healthiness (Pfeiffer et al., 2021; Yang et al., 2023). Apart from holistic aesthetic design, there are specific classifications of aesthetics. Classical aesthetics, a prevalent form of aesthetics, encompasses design principles consisting of balance, order, symmetry and pattern repetition (Hagen, 2021; Lavie and Tractinsky, 2004; Ye et al., 2020b). These features can capture consumers' attention and establish positive attribute associations in their minds (Loken, 2006; Ye et al., 2020b). For instance, Hagen (2021) designed food based on classical aesthetics characteristics and found that consumers perceived high classical aesthetic foods as healthier. Additionally, plating food in a classical aesthetic manner can also promote healthy judgment (Liu et al., 2023).

Packaging research has demonstrated that the visual symmetry can influence consumers' perception of product attributes and consumption decisions through aesthetics judgment. For instance, when the information items on the packaging were presented in a symmetrical form, consumers' aesthetics evaluation of packaging and their purchase intentions were enhanced (Bigoin-Gagnan and Lacoste-Badie, 2018). Velasco et al. (2019) found a stronger association between symmetrical design and approach motivation compared to asymmetrical design. Moreover, Romeo-Arroyo et al. (2023) provided empirical evidence that consumers positively associated symmetrical packaging design with premiumness. These findings suggest that consumers exhibit more preferences towards both foods and packaging featuring high classical aesthetic characteristics.

Product images on the packaging can effectively convey information regarding food attributes. Prior studies have revealed that the locations of food images on packaging can significantly impact consumers' perceptions of flavor and calories, subsequently influencing their preferences for healthier food (Manippa et al., 2022; Togawa et al., 2019). Therefore, we assume that consumers also can evaluate the food healthiness based on the presentation of food images. Classical aesthetics is linked to the perception of utilitarian properties (Liu et al., 2023; Ye et al., 2020b). Given that healthiness is a utilitarian attribute of food and the classical aesthetics of the food designs can affect the healthiness perception (Hagen, 2021; Liu et al., 2023), we hypothesized that this influence could extend to food images on packaging. Specifically, we assumed that when all other packaging elements remained consistent across conditions, presenting food images on the packaging in a high classical aesthetic style would elicit higher perceived healthiness.

H1. High classical aesthetic presentation of product images on the packaging (versus low classical aesthetics) elicits higher perceived healthiness.

2.3. The mediating role of perceived naturalness

There has been an intensified pursuit of sustainable and healthy living among individuals, resulting in an increasing prioritization of the natural attribute of food (Kumar et al., 2021). The naturalness of food, which encompasses notions of being natural, pure, and unprocessed, is a positive attribute that can influence consumers' attitude and acceptance towards food (Hagen, 2021; Rozin, 2005). Prior research indicates that people exhibit a strong favoring towards natural foods due to their specific advantages in terms of promoting health and being environmentally friendly (Rozin, 2005; Rozin et al., 2004), suggesting a lay belief of "natural = healthy" (Binninger, 2017; Wang et al., 2022).

The naturalness of food can be effectively communicated to consumers through packaging designs. For instance, the utilization of matte materials for packaging can enhance the perception of food naturalness (Marckhgott and Kamleitner, 2019). Moreover, incorporating familiar common names in the ingredient information on the packaging can augment consumers' perception of food naturalness (Chambers et al., 2018). Minimalist packaging can also lead consumers to perceive food as more natural (Saintives and Meral, 2024).

Hagen (2021) highlighted the association between classical aesthetics and the sense of naturalness, emphasizing that approximate symmetry, order and balance are prevalent in nature. For instance, up to 99% of animals exhibit approximate symmetry (Finnerty, 2005), self-similarity patterns in plants exhibit order (Mandelbrot, 1982), and the golden spiral arrangement of plant leaves exhibits balance (Mitchison, 1977). These aesthetic patterns form the foundation for proposing that classical aesthetic design of food has the potential to induce naturalness perception and subsequently promote perceived healthiness. Similarly, a recent study found that green products incorporating classical aesthetic features are perceived to possess greater naturalness (He et al., 2024).

Accordingly, we hypothesized that the perceived naturalness plays a mediating role between classical aesthetic presentation of food images and perceived healthiness.

H2. Compared to low classical aesthetic presentation, high classical aesthetic presentation of food images on packaging enhances perceived naturalness, thereby increasing perceived healthiness.

2.4. The moderating role of exhibiting organic label

Food firms often apply various labels on packaging to effectively convey unobtainable information regarding product quality and attributes to consumers. For instance, nutrition labels or health labels on packaging can facilitate communication about food health information and are effective in guiding healthy choices (Hagmann and Siegrist, 2020; Maesen et al., 2022).

Foods labeled as organic are often perceived as being free of "chemicals" and "growth hormones" and are considered to be inherently "natural" (Davies et al., 1995), aligning with the concept of perceived naturalness to a certain extent. Prior research has demonstrated that packaging featuring organic labels or naturalness claims can effectively communicate the natural attributes of food and influence subsequent consumers' behaviors (Binninger, 2017; Donato and Adıgüzel, 2022; Richetin et al., 2022; Wang et al., 2022), especially when the labels originate from an independent authority (Lunardo and Saintives, 2013). For instance, Binninger (2017) found that foods packaged with the European organic label were perceived as more natural compared to those packaged with only pictures and green colors. In our study, we used the organic label that is an official certification for products produced, processed and sold in accordance with national standards. This certified label is likely to capture consumers' attention and enhance their reliance on organic labels when making judgments about food (Gorton et al., 2021). Therefore, the presence of organic labels on food packaging signifies a connection with nature while potentially impeding high

classical aesthetics from evoking the perception of naturalness.

Building on the aforementioned discussion, we hypothesized that presenting the organic label on the packaging would moderate the naturalness-mediated effect between the aesthetic food image presentation and perceived healthiness (see Fig. 1 for the conceptual framework).

H3. When the organic label is present on food packaging (versus absent), the mediating effect of naturalness is attenuated.

3. Pilot study

The pilot study was conducted to gather preliminary evidence regarding the link between the classical aesthetic presentation (i.e., symmetrical, orderly and balanced) of food images on packaging and perception of healthiness. To this end, we asked the participants to search real-world food packaging and evaluate their classical aesthetic design and the perceived healthiness of the corresponding foods.

3.1. Method

We conducted a 2 (aesthetic presentation: high vs. low) × 6 (food type) between-subjects experiment. The participants were randomized to search for and upload a real-world packaging picture of one of six snack categories (Candy & Chocolate, Cookies & Crackers, Chips & Crisps, Nuts & Dried fruit, Snack cakes & Pastries, and Jerky), featuring a high or low classical aesthetic presentation of the food images on the front of packaging (see Appendix A for specific requirements). These six categories of snacks were determined based on the classification of snack foods available on prominent retail platforms such as Amazon and Walmart. We calculated the necessary sample size using G*Power 3.1 (*power* = 0.80, $f = 0.25$, $\alpha = 0.05$), which required at least 212 participants. Therefore, we recruited 347 participants (72.33% female; $M_{age} = 30.13$, $SD = 7.06$) on the Credamo platform (https://www.credamo.com/), a professional online data collection platform in China that is extensively used by researchers across various disciplines for conducting experiments (Liu et al., 2024; Xu et al., 2024). The participant pool within the Credamo platform is broadly distributed nationwide and encompasses a diverse range of backgrounds (see Appendix A for geographical distribution of recruited participants in pilot study).

After uploading the packaging pictures, participants evaluated their perceptions of food's healthiness and classical aesthetics of the packaging on a 7-point scale (Hagen, 2021). Finally, participants provided their demographic information.

3.2. Measurement

Perceived healthiness. Participants rated the healthiness of the food using a five-item scale that included perceptions of being "healthy," "nutritious," "good for me," "low in fat," and "low in calories" ($\alpha = 0.91$).

Classical aesthetics. Participants assessed the degree of symmetry, order, and balance in the presentation of food images on the packaging ($\alpha = 0.94$).

3.3. Results

The results of reliability and validity tests for the perceived healthiness and classical aesthetics are listed in Appendix F. Specifically, both Cronbach's α and composite reliability (CR) values exceeded the recommended threshold of 0.70 (Hair et al., 2022), and the average variance extracted (AVE) values were greater than the threshold of 0.50 (Sarstedt et al., 2022), which indicated satisfactory reliability and validity.

Manipulation check for classical aesthetics. Two-way between-subject analysis of variance with aesthetics and food type factors was conducted. The ANOVA results showed that aesthetic rating was significantly higher when participants searched and evaluated the packaging featuring food images in a high classical aesthetic style ($M_{high} = 5.81$, $SD = 0.59$) compared to those in a low style ($M_{low} = 2.34$, $SD = 0.91$; $F(1, 335) = 1811.05$, $p < 0.001$, $\eta 2\ p = 0.84$). There was no significant main effect of food type on aesthetic rating ($F(5, 335) = 1.95$, $p = 0.09$, $\eta 2\ p = 0.03$), and the interaction effect was also not significant ($F(5, 335) = 0.89$, $p = 0.49$, $\eta 2\ p = 0.01$). The manipulation check was successful for the six food categories (all $ps < 0.001$, see Appendix A for details).

Perceived healthiness. The ANOVA results showed that the main effects of aesthetics ($F(1, 335) = 32.50$, $p < 0.001$, $\eta 2\ p = 0.09$) and food type ($F(5, 335) = 30.82$, $p < 0.001$, $\eta 2\ p = 0.32$) were both significant. We did not observe a significant interaction effect ($F(5, 335) = 0.29$, $p = 0.92$, $\eta 2\ p = 0.004$). Participants exhibited a significantly higher healthiness perception for high classical aesthetic packaging ($M_{high} = 4.72$, $SD = 1.24$) than low ones ($M_{low} = 4.06$, $SD = 1.32$). This trend was consistent across all six food categories (all $ps < 0.05$, see Appendix A for details).

3.4. Discussion

The pilot study, using real-world food packaging, demonstrated the significant positive correlation between the classical aesthetic presentation of food images and healthiness perception, thereby supporting Hypothesis 1. Subsequently, we conducted further validation of our hypotheses in a more controlled setting.

4. Study 1

Study 1 conducted two experiments to investigate whether the classical aesthetic presentation of food images on packaging would influence consumers' perceived healthiness and explored the naturalness-mediated role.

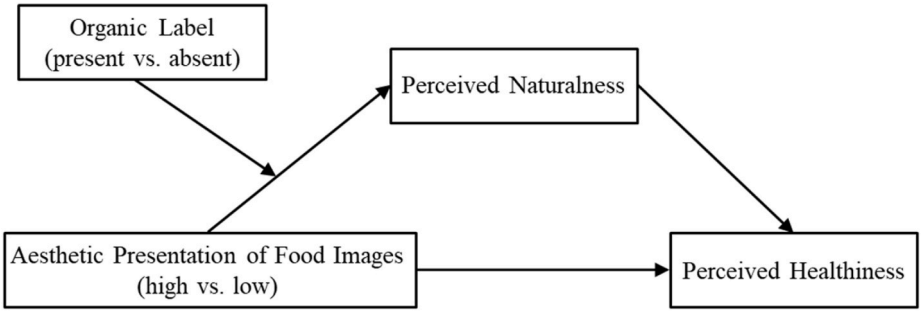


Fig. 1. Conceptual framework.

4.1. Study 1a

4.1.1. Pretest for stimuli

Study 1a used the mushroom crisp as the food stimuli and created two packaging images according to the classical aesthetics definition (Hagen, 2021). Specifically, in high aesthetics condition, the mushroom crisp images on the packaging were presented in symmetrical, orderly and balanced style, while in low aesthetics condition, the images were presented in asymmetrical, disorderly and imbalanced style. All other elements on the packaging were kept the same (see Appendix B for the packaging). We recruited 80 participants from Credamo platform. These participants were assigned to evaluate the packaging with high aesthetics design or low design randomly, specifically focusing on the symmetry, order and balance of the displayed food images, using the same 7-point scale employed in the pilot study ($\alpha = 0.92$). The *t*-test result indicated a significant difference in the aesthetic ratings between these two packaging images ($M_{high} = 6.20$, $SD = 0.52$; $M_{low} = 4.48$, $SD = 1.97$; $t(78) = -5.32$, $p < 0.001$, $d = 1.19$).

4.1.2. Method

We conducted a single factor (aesthetic presentation: high vs. low) between-subjects experiment. Based on the minimum required sample size of 128 calculated by G*Power 3.1 (*power* = 0.80, $d = 0.50$, $\alpha = 0.05$), we recruited 170 participants (71.76% female; $M_{age} = 30.62$, $SD = 6.76$; detailed demographic characteristics are provided in Appendix B) through Credamo platform. The participants were instructed to mentally simulate the process of purchasing the mushroom crisp. Then they viewed the mushroom crisp packaging either with high or low aesthetics design randomly, followed by the ratings on perceived healthiness, naturalness, tastiness and freshness of the food using 7-point scales which were adopted in a prior study (Hagen, 2021). Finally, the participants completed the questions assessing the aesthetics manipulating check, attention check and reported demographic information.

4.1.3. Measurement

Perceived healthiness. The measurements for perceived healthiness used the identical five-item scale employed in the pilot study ($\alpha = 0.85$).

Perceived naturalness. Participants rated the naturalness of the mushroom crisp with "natural" and "unprocessed" ($\alpha = 0.78$).

Perceived tastiness. Participants rated the tastiness of mushroom crisp with "tasty" and "delicious" ($\alpha = 0.85$).

Perceived freshness. Participants reported how fresh the mushroom crisp they perceived.

Classical aesthetics. The manipulation test for aesthetics used the same three items as in pretest ($\alpha = 0.90$).

4.1.4. Results

The results of Cronbach's α, CR and AVE of the measurement scales indicated satisfactory levels of reliability and validity (see Appendix F for details).

Manipulation check for classical aesthetics. The participants perceived the presentation of mushroom crisp images as highly orderly, symmetrical and balanced in high classical aesthetics condition ($M_{high} = 6.20$, $SD = 0.49$) compared to low aesthetics condition ($M_{low} = 3.76$, $SD = 1.52$; $t(168) = -13.99$, $p < 0.001$, $d = 2.16$), indicating the successful manipulation.

Perceived healthiness. Participants exhibited a significantly higher healthiness rating of the mushroom crisp in high classical aesthetics condition ($M_{high} = 5.91$, $SD = 0.82$) than in low condition ($M_{low} = 5.57$, $SD = 0.96$; $t(168) = -2.49$, $p = 0.01$, $d = 0.38$).

Perceived naturalness. Participants exhibited a significantly higher naturalness perception of the mushroom crisp in high classical aesthetics condition ($M_{high} = 5.76$, $SD = 0.91$) than in low condition ($M_{low} = 5.26$, $SD = 1.44$; $t(168) = -2.73$, $p = 0.007$, $d = 0.42$).

Equality check for perceived tastiness and freshness. The mushroom crisp in high and low aesthetic packaging were rated as equally tasty ($M_{high} = 5.68$, $SD = 1.02$; $M_{low} = 5.45$, $SD = 1.23$; $t(168) = -1.32$, $p = 0.19$, $d = 0.20$) and fresh ($M_{high} = 5.36$, $SD = 1.21$; $M_{low} = 5.08$, $SD = 1.71$; $t(168) = -1.24$, $p = 0.22$, $d = 0.19$). Therefore, perceived tastiness and freshness were ruled out as alternative explanations.

Mediation. Simple mediation with PROCESS Model 4 (95% confidence intervals; 5000 samples) revealed that perceived naturalness served as a mediator between aesthetic presentation and perceived healthiness (*indirect effect* = 0.28, $SE = 0.09$, 95% CI = [0.09, 0.46]; as shown in Fig. 2).

4.2. Study 1b

To bolster the robustness and generalizability of above findings, we conducted study 1b that employed another type of food and designed the presentation of food images in a different way.

4.2.1. Pretest for stimuli

In line with study 1a, two packaging images of cracker were created to respectively exemplify high and low classical aesthetics designs (see Appendix C). We recruited one hundred participants on the Credamo platform. They were randomized to evaluate either high or low classical aesthetic cracker packaging using the identical items in study 1a ($\alpha = 0.88$). T-test results indicated a clear disparity in the aesthetic ratings between the two packaging images ($M_{high} = 5.61$, $SD = 0.77$; $M_{low} = 2.90$, $SD = 1.39$; $t(98) = 12.04$, $p < 0.001$, $d = 2.41$).

4.2.2. Method

We conducted a single factor (aesthetic presentation: high vs. low) between-subjects experiment through Prolific platform (https://www.prolific.com/), which is a UK-based online platform for researchers to collected data with a diverse global participant pool. We recruited 150 USA participants (56.00% female; $M_{age} = 38.79$, $SD = 10.93$, see Appendix C for detailed demographic characteristics), exceeding the recommended sample size of 128 as estimated by the G*Power 3.1 with the same setting as study 1a. They were randomized to evaluate cracker packaging with high or low classical aesthetics design. The procedure of the task and the measurements of perceived healthiness ($\alpha = 0.91$), naturalness ($\alpha = 0.82$), tastiness ($\alpha = 0.93$) and freshness were the same with study 1a. Finally, participants finished the manipulation check for aesthetics ($\alpha = 0.87$), attention check and reported the demographic information.

4.2.3. Results

The results of Cronbach's α, CR and AVE of the measurement scales also demonstrated satisfactory levels of reliability and validity (see Appendix F for details).

Manipulation check for classical aesthetics. Participants exhibited a higher aesthetic rating when the images were presented in high classical aesthetic style ($M_{high} = 5.49$, $SD = 1.00$) than that in low style ($M_{low} = 4.74$, $SD = 1.46$; $t(148) = -3.67$, $p < 0.001$, $d = 0.60$), indicating the successful manipulation.

Perceived healthiness. Participants exhibited an enhanced healthiness perception of the cracker in high classical aesthetic style ($M_{high} = 4.02$, $SD = 1.15$) than in low style ($M_{low} = 3.40$, $SD = 1.22$; $t(148) = -3.19$, $p = 0.002$, $d = 0.52$).

Perceived naturalness. Participants exhibited an enhanced naturalness perception of the cracker in high classical aesthetic style ($M_{high} = 3.72$, $SD = 1.31$) than in low style ($M_{low} = 3.03$, $SD = 1.34$; $t(148) = -3.20$, $p = 0.002$, $d = 0.52$).

Equality checks for perceived tastiness and freshness. The cracker in high and low aesthetic packaging were evaluated as equally tasty ($M_{high} = 4.53$, $SD = 1.42$; $M_{low} = 4.47$, $SD = 1.46$; $t(148) = -0.26$, $p = 0.80$, $d = 0.04$) and fresh ($M_{high} = 4.68$, $SD = 1.23$; $M_{low} = 4.31$, SD

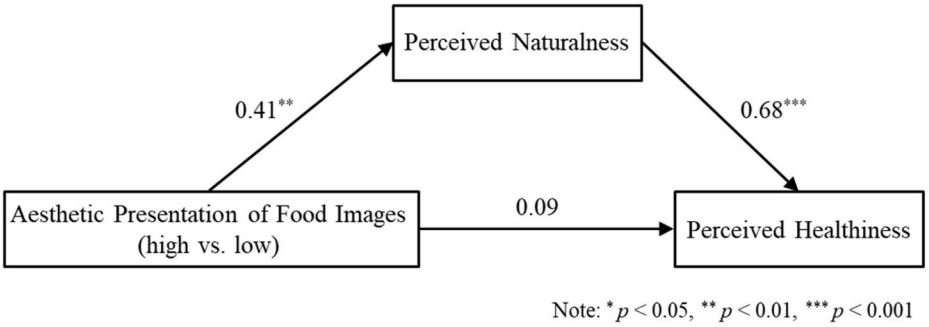


Fig. 2. Mediation path diagram (Study 1a).

= 1.51; $t(148) = -1.66$, $p = 0.10$, $d = 0.27$). Therefore, perceived tastiness and freshness are not the viable alternative explanations.

Mediation. Simple mediation with PROCESS Model 4 (95% confidence intervals; 5000 samples) revealed that perceived naturalness played a mediating role between aesthetic presentation and perceived healthiness (*indirect effect* = 0.32, *SE* = 0.10, 95% CI = [0.12, 0.53]; see Fig. 3).

4.3. Discussion

Consistent with our predictions, study 1a and 1b used broad samples and different foods to indicate that consumer regarded the food in the packaging exhibiting the images in high classical aesthetic style as healthier (H1). This effect can be attributed to the perceived naturalness of such food (H2). Moreover, the results showed that neither freshness nor tastiness was influenced by the aesthetic presentation.

5. Study 2

Study 2 aimed to further validate the aesthetics effect on perceived healthiness and mediating mechanism of naturalness. Besides, we also tested this aesthetics effect on the subsequent purchase intention.

5.1. Method

We conducted a single factor (aesthetic presentation: high vs. low) between-subjects experiment and calculated the necessary sample size using G*Power 3.1 (*power* = 0.80, $d = 0.50$, $\alpha = 0.05$), which required at least 128 participants. Therefore, we recruited 200 participants (64.50% female; $M_{age} = 31.30$, $SD = 7.65$; see Appendix D for detailed demographic characteristics) on Credamo platform. They were randomly assigned to evaluate cracker packaging featuring high or low classical aesthetics, which were identical to those of study 1b. The procedure of the task and the measurements of perceived healthiness ($\alpha = 0.83$), naturalness ($\alpha = 0.78$), tastiness ($\alpha = 0.84$) and freshness were identical to those in study 1a & 1b. Participants completed the manipulation check for aesthetics ($\alpha = 0.83$), attention check and reported the demographic information in the final stage.

5.2. Results

The results of Cronbach's α, CR and AVE of the measurement scales indicated satisfactory levels of reliability and validity as study 1 (see Appendix F for details).

Manipulation check for classical aesthetics. Participants exhibited a higher aesthetic rating when the images were presented in high classical aesthetic style ($M_{high} = 6.00$, $SD = 0.56$) than in low style ($M_{low} = 3.90$, $SD = 1.43$; $t(198) = -13.67$, $p < 0.001$, $d = 1.93$), indicating the successful manipulation.

Perceived healthiness. Participants demonstrated a significantly higher healthiness rating for the cracker presented in high classical aesthetic style ($M_{high} = 5.44$, $SD = 0.73$) than in low style ($M_{low} = 5.02$, $SD = 1.05$; $t(198) = -3.22$, $p = 0.002$, $d = 0.46$).

Perceived naturalness. Participants exhibited a significantly higher naturalness perception of the cracker presented in high classical aesthetic style ($M_{high} = 5.25$, $SD = 1.02$) than in low style ($M_{low} = 4.68$, $SD = 1.38$; $t(198) = -3.36$, $p = 0.001$, $d = 0.47$).

Purchase intention. Participants had a stronger purchase intention for high aesthetic packaging ($M_{high} = 5.54$, $SD = 0.80$) than low one ($M_{low} = 5.23$, $SD = 1.13$; $t(198) = -2.25$, $p = 0.03$, $d = 0.32$; see Fig. 4).

Equality checks for perceived tastiness and freshness. The cracker in high and low aesthetic packaging were rated as equally tasty ($M_{high} = 5.62$, $SD = 0.86$; $M_{low} = 5.59$, $SD = 1.23$; $t(198) = -0.20$, $p = 0.84$, $d = 0.03$) and fresh ($M_{high} = 5.71$, $SD = 0.94$; $M_{low} = 5.60$, $SD = 1.28$; $t(198) = -0.69$, $p = 0.49$, $d = 0.10$). Therefore, perceived tastiness and freshness are not the viable alternative explanations.

Mediation. Simple mediation with PROCESS Model 4 (95% confidence intervals; 5000 samples) revealed that perceived naturalness served as a mediator between aesthetic presentation and perceived healthiness (*indirect effect* = 0.33, *SE* = 0.10, 95% CI = [0.14, 0.53]).

Serial Mediation. Serial mediation analysis using PROCESS Model 6 (95% confidence intervals; 5000 samples) identified the significant

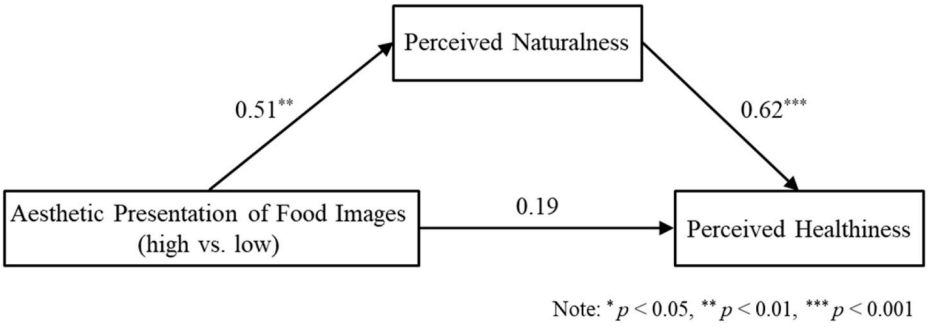


Fig. 3. Mediation path diagram (Study 1b).

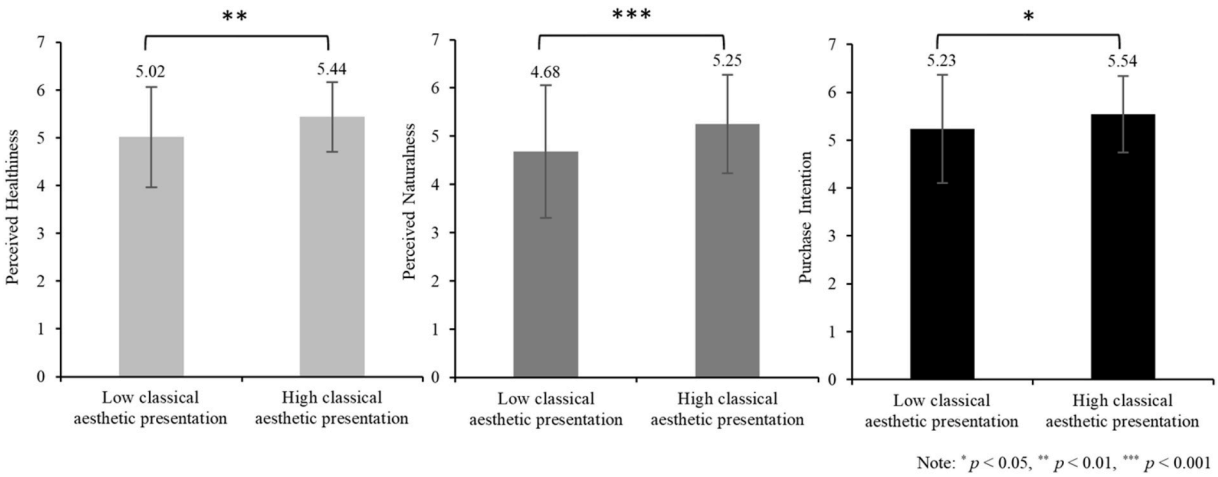


Fig. 4. Effect of classic aesthetic presentation on perceived healthiness, naturalness and purchase intention (Study 2).

indirect effect of classical aesthetic presentation on purchase intention, mediated sequentially by perceived naturalness and healthiness (*indirect effect* = 0.13, *SE* = 0.06, 95% CI = [0.04, 0.25]; see Fig. 5).

5.3. Discussion

Study 2 replicated the findings of study 1, indicating that the classical aesthetic presentation of food images on packaging can positively affect consumers' perceived healthiness, which was mediated by the sense of naturalness. What is more, consumers showed higher purchase intentions when the images on the packaging were presented in high classical aesthetic style, and this positive effect was sequentially influenced by the perceived naturalness and healthiness.

6. Study 3

Study 3 aimed to examine whether presenting the organic label on food packaging moderated the effect of classical aesthetic presentation on the perceived healthiness as well as the naturalness-mediated effect.

6.1. Pretest for stimuli

Study 3 used the strawberry crisp as food stimuli and a similar pretest was conducted to confirm the high and low aesthetic packaging (see Appendix E). 128 participants were recruited from Credamo platform and evaluated packaging using same three items of aesthetics in study 1 (α = 0.92). T-test results indicated a clear disparity in the aesthetic degree between the two packaging images (M_{high} = 6.11, *SD* = 0.50; M_{low} = 3.81, *SD* = 1.73; $t(126) = 10.12$, $p < 0.001$, $d = 1.82$).

6.2. Method

A between-subjects experiment with two factors: aesthetic presentation (high vs. low) and organic label (present vs. absent) was conducted. Based on the minimum required sample size of 128 calculated by G*Power 3.1 (*power* = 0.80, $f = 0.25$, $\alpha = 0.05$), we recruited 230 participants (57.39% female; M_{age} = 31.89, *SD* = 9.97; detailed demographic characteristics are provided in Appendix E) from Credamo platform. They were randomly assigned to one of these four conditions.

The procedure for the organic label absent condition remained consistent with study 1&2. In the organic label present condition, participants were additionally showed an explicit description of the organic label: "The label affixed to the product packaging signifies official certification indicating compliance with national standards governing production, processing and sales of the products." Besides, the packaging they observed featured the label positioned in the upper left corner. Then participants moved on to rate the perceived healthiness (α = 0.79), naturalness (natural, unprocessed, pure; α = 0.73), tastiness (α = 0.73) and freshness that were used in study 1&2. Finally, participants complete the manipulation check for aesthetics (α = 0.91), attention check and reported the demographic information.

6.3. Results

The results of Cronbach's α, CR and AVE of the measurement scales indicated the satisfactory reliability and validity as our prior two studies (see Appendix F for details).

Manipulation check for classical aesthetics. Two-way between-subject analysis of variance with aesthetics and label factors was

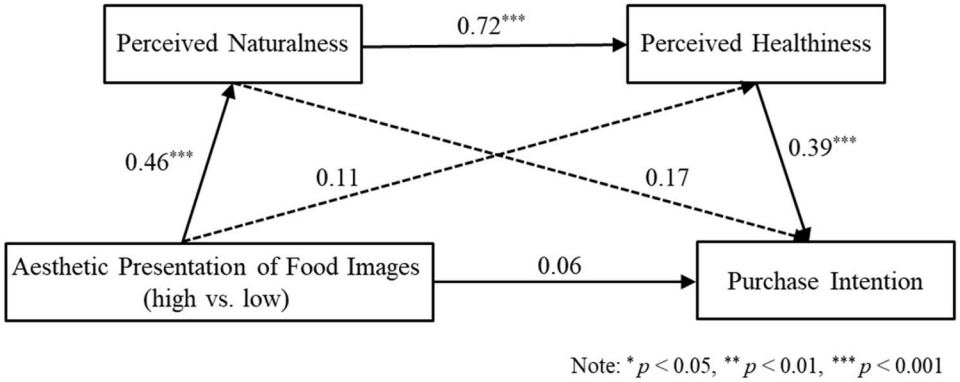


Fig. 5. Serial mediation path diagram (Study 2).

conducted. The results showed that aesthetics rating was significantly higher when the strawberry crisp images were presented in the high classical aesthetic style than in low style ($M_{high} = 6.28$, $SD = 0.51$; $M_{low} = 3.88$, $SD = 1.38$; $F(1, 226) = 300.13$, $p < 0.001$, $\eta 2\ p = 0.57$). The effect of organic label on classical aesthetics did not reach significance ($F(1, 226) = 3.10$, $p = 0.08$, $\eta 2\ p = 0.01$). In addition, the interaction effect was also not significant ($F(1, 226) = 1.49$, $p = 0.22$, $\eta 2\ p < 0.01$). Therefore, the manipulation was successful.

Perceived healthiness. The ANOVA results indicated that the interaction effect between aesthetic presentation and organic label on perceived healthiness was significant ($F(1, 226) = 5.03$, $p = 0.03$, $\eta 2\ p = 0.02$). As shown in Fig. 6, simple effect analysis indicated that when the organic label was present, participants showed no different perceptions of healthiness in the low and high classical aesthetic styles ($M_{high} = 5.95$, $SD = 0.65$; $M_{low} = 5.89$, $SD = 0.62$; $p = 0.68$). However, when the organic label was absent, participants exhibited a significantly higher healthiness perception of the strawberry crisp in high classical aesthetic style ($M_{high} = 5.73$, $SD = 0.59$) than low style ($M_{low} = 5.24$, $SD = 1.04$; $p < 0.001$).

Perceived naturalness. We also observed a significant interaction effect on perceived naturalness ($F(1, 226) = 5.25$, $p = 0.02$, $\eta 2\ p = 0.02$). As shown in Fig. 7, simple effect analysis confirmed that participants show no different perceptions of naturalness in the high and low classical aesthetic styles when exhibiting the organic label on packaging ($M_{high} = 5.94$, $SD = 0.77$; $M_{low} = 6.03$, $SD = 0.74$; $p = 0.57$). However, without the organic label, participants demonstrated a significantly higher naturalness perception of the strawberry crisp in high classical aesthetic style ($M_{high} = 5.80$, $SD = 0.65$) than in low style ($M_{low} = 5.40$, $SD = 1.02$; $p = 0.008$).

Equality checks for perceived tastiness and freshness. Although a significant interaction effect on perceived tastiness was observed ($F(1, 226) = 4.45$, $p = 0.04$, $\eta 2\ p = 0.02$), simple effect analysis indicated that the strawberry crisps in high or low aesthetics packaging were rated as equally tasty regardless of the presence or absence of the organic label (label present: $M_{high} = 5.89$, $SD = 0.74$; $M_{low} = 6.10$, $SD = 0.87$, $p = 0.18$; label absent: $M_{high} = 5.91$, $SD = 0.71$; $M_{low} = 5.66$, $SD = 0.99$, $p = 0.10$). Besides, we did not observe the significant interaction effect on perceived freshness ($F(1, 226) = 1.23$, $p = 0.27$, $\eta 2\ p < 0.01$).

Moderated mediation. A moderated mediation analysis using Model 7 (95% confidence intervals; 5000 samples) indicated that in the organic label present condition (coded as 1), perceived naturalness did not serve as a mediator between classical aesthetic presentation and perceived healthiness (*indirect effect* = −0.06, *SE* = 0.09; 95% CI = [−0.24, 0.13]). Conversely, in the organic label absent condition (coded as 0), perceived naturalness played a mediating role (*indirect effect* = 0.27, *SE* = 0.11; 95% CI = [0.06, 0.51]; see Fig. 8).

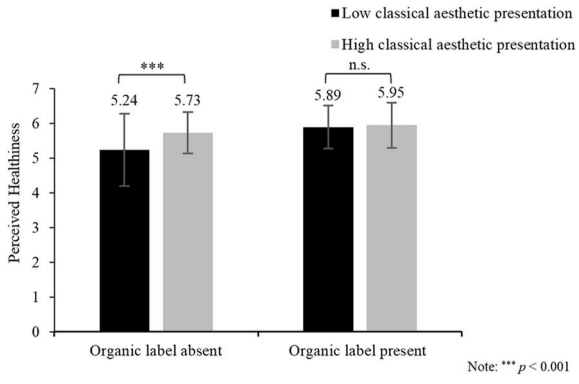


Fig. 6. The interaction effect of aesthetic presentation and organic label on perceived healthiness (Study 3).

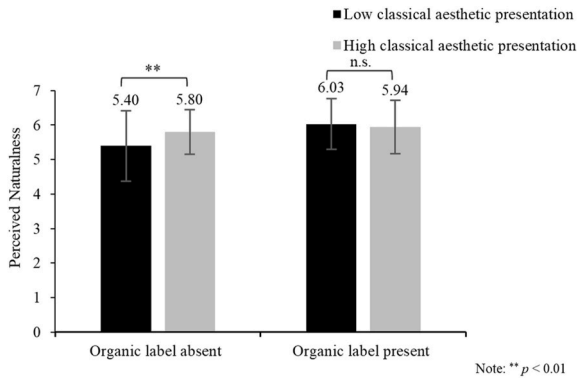


Fig. 7. The interaction effect of aesthetic presentation and organic label on perceived naturalness (Study 3).

6.4. Discussion

In accordance with Hypothesis 3, the inclusion of the organic label moderated the mediating role of perceived naturalness. Specifically, under the organic label present condition, perceived naturalness did not act as a mediator.

7. General discussion

7.1. Conclusions

The investigation of how consumers form judgments regarding product attributes through their sensory experience of packaging has garnered significant attention among researchers (Krishna et al., 2017). While prior research on food packaging has predominantly highlighted the influences of shape, size and color on consumer perceptions of tastiness, healthiness and other attributes, limited attention has been given to exploring the impact of presentation design for product images on food packaging in influencing attribute perception (Motoki and Togawa, 2022). Our study investigated the effect of classical aesthetic presentation of food images on perceived healthiness while uncovering the underlying mechanism and boundary associated with this aesthetics effect.

Specifically, the pilot study provided preliminary evidence through the real-world packaging of diverse foods to confirm the positive link between classical aesthetics presentation and the perception of healthiness. The study 1 and study 2 conducted controlled experiments, where the classical aesthetic design of packaging stimuli was manipulated, verifying that the utilization of high classical aesthetic style in presenting food images on packaging elicits a heightened perception of healthiness. This effect is mediated by the sense of naturalness: high classical aesthetic presentation of the food images leads consumers to perceive food as more natural, thereby enhancing healthiness perception. Moreover, the findings of study 2 additionally revealed that perceived naturalness and healthiness serially played the mediating roles between aesthetic presentation and purchase intention. Study 3 indicated that presenting an organic label on the packaging moderates the mediating role played by the perceived naturalness. Notably, presenting an organic label hinders the effect of aesthetics from evoking the perception of naturalness, thereby undermining the mediating effect.

7.2. Theoretical contributions

The study contributes to the existing literature on product image design in food packaging by expanding upon prior studies that have primarily concentrated on holistic packaging design elements (e.g., color, shape, material), and their impacts on healthiness judgments (Mai et al., 2016; Yarar et al., 2019; Ye et al., 2020a). However, it still lacks a

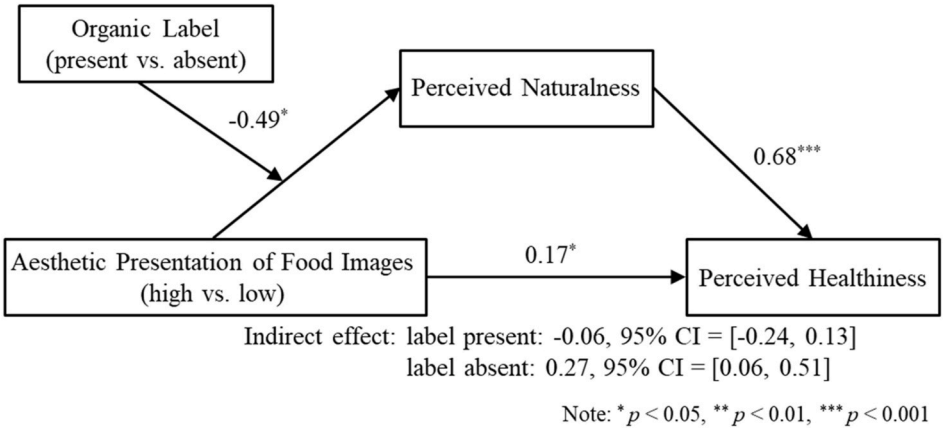


Fig. 8. Moderated mediation path diagram (Study 3).

comprehensive understanding regarding the influence of specific constituent elements of packaging, such as product images, on healthiness perception. While some studies have examined the content characteristics, size, amount and location of food images displayed on packaging in relation to consumers' attribute processing (Madzharov and Block, 2010; Mizutani et al., 2010; Petit et al., 2018; Togawa et al., 2019), they have overlooked the impact of food image presentation design on perceived healthiness. To address this gap, our study innovatively explores the effect of aesthetic food image presentation on consumers' judgement of healthiness and enhances our understanding of the roles played by product images on packaging in attributes perceptions. Furthermore, the finding of aesthetics effect of food images on healthiness perception contributes to the existing body of literature on the cross-modal influence, which highlights how stimuli in one sensory modality (vision, audition, gustation, touch, olfaction) can affect the perception of other modalities (Biswas et al., 2019; Biswas and Szocs, 2019; Spence, 2021). For instance, previous studies have demonstrated that food-related ambient scent (olfactory cue) and ambient music volume (auditory cue) can affect the gustatory consumption of the food (Biswas et al., 2019; Biswas and Szocs, 2019). Our study enriches this domain by showing cross-modal effect of the visual aesthetic cue on food healthiness perception and purchase intention.

Secondly, this study extends the application of aesthetics theory in food packaging research by bridging the existing knowledge gap regarding the correlation between aesthetic packaging and food attributes. Prior literature has primarily explored the influences of holistic aesthetics on neuropsychological and behavioral responses of the consumers (Hubert et al., 2013; Reimann et al., 2010). Building on these research, our study expands upon the findings that "classical aesthetics can promote healthiness perception" (Hagen, 2021) by uncovering the aesthetics effect of presenting food on packaging, thereby enriching classical aesthetics theory. While prior evidence has demonstrated how classical aesthetic features of food influence healthiness perception through a sense of naturalness, this mechanism has not been verified within a packaging context. Therefore, our study not only broadens our understanding from focusing solely on food itself to considering its presentation but also provides novel evidence that classical aesthetic design in packaging can convey naturalness of food and enhance healthiness perception. Consequently, it further enhances the practical applicability of classical aesthetics theory.

Finally, this study contributes a novel factor that can attenuate the association between aesthetic presentation and naturalness, thereby expanding the scope of packaging aesthetics effect. Previous research has primarily identified moderating factors such as brand prominence (Cesareo et al., 2022) and product type (Sundar et al., 2020) in relation to aesthetics effect. Our study introduces the organic label used in the food market as a moderating variable. Prior study indicated that showing the reminder that the food has undergone artificial modification could alleviate the mediating effect of naturalness (Hagen, 2021). In line with this finding, our results revealed that presenting an organic label on packaging can also disrupt the effect of classical aesthetics on naturalness perception. This research establishes a new boundary for applying classical aesthetics theory in packaging studies and provides valuable insights into label effects within packaging research.

7.3. Managerial implications

The consumption of food has consistently remained the primary focal point in marketing practice, as there is an emerging inclination among consumers towards an escalating concern for maintaining healthy diets. Moreover, according to the industry analysis, the food packaging market is continually expanding and is expected to reach 512 billion dollars by 2028 (Statista, 2024). Food packaging plays a crucial role in fostering healthy choices, reducing food waste, and advancing sustainability efforts. Therefore, it holds practical significance to enhance consumers' healthiness perception of food through effective packaging design, and facilitate purchasing decisions for healthier food. Our study provides some practical implications.

First, it provides valuable insights for food firm to effectively convey the healthiness of their products in a more convenient manner. Specifically, packaging designers should incorporate classical aesthetic principles (orderly, symmetrical and balanced) when presenting food images on packaging. Second, if there are visual cues certifying the food as natural, organic or healthy, such as the organic label, it is more crucial to prioritize highlighting these visual cues on the packaging. Third, our study provides novel insights to assist consumers in judging healthiness and avoiding potential deception based solely on the aesthetics of food packaging. While classical aesthetic presentation of food images may signal healthiness, it can also mislead consumers into perceiving an "unhealthy" product as being "healthy". Therefore, based on the marketing ethics principles of honesty and transparency and the normative approach of developing guidelines (Friedmann et al., 2024), we addressed this misleading ethical issue from three perspectives. First, for consumers with specific health goals, it is advisable not to solely rely on the classical aesthetic design as a potential indicator; instead, they should also consider information such as nutrition facts tables and labels as reliable diagnostics. Second, marketers should adopt a consumer-oriented relational view by disclosing that the classical aesthetic presentation of food images featured on packaging has undergone artificial enhancement. Such declaration serves to demonstrate transparency and prominently highlight the ethical considerations underlying the use of this kind design of images. Finally, policymakers should enhance the utilization of food labels to communicate health information, thereby eliminating the effort for consumers to integrate

various pieces of information to judge the food healthiness. Additionally, policymakers can launch educational campaigns to uncover hidden biases under the halo effect of aesthetics on consumers' perceptions, and educate the public on the implications of food labels, thereby facilitating more informed healthier choices.

7.4. Limitations and future research

Our study possesses several limitations that necessitate further enhancements in future research. Firstly, the samples of our study were solely recruited from online platforms including Credamo and Prolific. This convenience sampling approach may potentially result in the samples not fully representing a broader population, consequently limiting the generalizability of our findings. Therefore, the inclusion of more diverse samples from various recruitment channels and actual consumers through field experiments in future research is recommended to enhance the generalizability and external validity of the findings. Specifically, increasing sample size can ensure broader representation of the target population. This can be achieved by utilizing multiple sources including additional online platforms (e.g., professional platforms like Amazon Mechanical Turk, social media) or offline scenes (e.g., retail environments) to access a wider range of participants from diverse cultural and demographic backgrounds. During recruitment, rigorous sampling strategies should be employed to ensure both the independence and randomization of the sample selection process. Besides, since our participants were recruited through the online platforms, the effects of their digital environment proficiency should be considered. For example, participants' digital literacy may influence their responses to the visual packaging stimuli. In our pilot study, we examined this factor and found that neither the evaluations of packaging stimuli nor the subsequent aesthetic effects on healthiness perception were influenced by digital literacy (see Appendix A for detailed results). However, future research should still consider measuring this individual difference and examining its influence on the consumers' responses in various online experimental contexts. For instance, a recent study demonstrated that digital literacy affected the green consumption behavior (Hu and Meng, 2023). Therefore, when conducting research on sustainable packaging or other related topics, we ought to factor in individual differences in digital literacy.

Secondly, this study primarily employed packaging pictures as stimuli to investigate the impact of aesthetic image presentation on consumers' perceived healthiness of the food, as many marketing researchers did in their work investigating packaging effects on food perception and decision (Togawa et al., 2019; Ye et al., 2020a). However, there may exist a certain disparity between these artificial stimuli and the actual packaging encountered during consumption scenarios, thereby potentially limiting our ability to fully capture consumer responses in authentic shopping situations. In our pilot study, we investigated the responses of participants to the real-world packaging stimuli, providing initial evidence of the association between classical aesthetic design and perceived healthiness "in the wild". This enhances the applied relevance of our findings to some extent. However, future research should consider utilizing physical food packaging to examine the effect of aesthetic design of packaging within both laboratory and real shopping environments (e.g., in-store) by conducting field experiments. Such studies can accurately capture consumer behavior in natural settings, allowing for causal inferences and thus ensuring high ecological and external validity by clarifying real-world applied contexts (Malodia et al., 2023). For example, conducting field experiments in supermarkets or snack food stores, as done in previous studies (Biswas and Szocs, 2019; Huang and Labroo, 2020; Ye et al., 2020a), can provide valuable insights into how packaging influences consumers. Besides, we can also collect the real data reflecting the consumers' responses to the actual packaging stimuli. For instance, some studies gathered packaging-related data (e.g., sales volume, prices, descriptions and elements number on packaging) from large retail platforms to explore the correlations between packaging designs and consumers' willingness to pay (Li et al., 2024; Ton et al., 2024). Furthermore, as the use of virtual reality technique has been progressively growing in marketing research (Hilken et al., 2022), we can also simulate an in-store environment via VR, and observe the evaluations and behaviors of consumers in response to different types of packaging. Furthermore, it is important to acknowledge that artificial designs of packaging stimuli may lead to limited familiarity with the packaging itself, raising concerns about participants' varying levels of familiarity with stimuli such as packaging and organic labels that might influence their perceptions of food. Hence, it is imperative that future studies consider this factor as a control variable.

Thirdly, our study solely focused on classical aesthetics while neglecting other types such as expressive aesthetics. Future research can explore the influence of alternative aesthetic presentations of product images on the healthiness perceptions to validate and broaden the applicability of our findings. Fourthly, our study has identified the moderating role of exhibiting organic label. Future research can investigate additional moderating factors, such as the utilization of alternative labeling methods and educational interventions to enhance consumer awareness, which would not only expand the understanding of boundary conditions related to the aesthetics effect but also guide the development of marketing strategies to address ethical concerns associated with aesthetic designs. Finally, this study mainly explores how packaging aesthetic design affects consumers' healthiness perception and only reveals its subsequent effect on purchase intention. Further research is needed to delve into more downstream effects (e.g., real consumption).

CRediT authorship contribution statement

Linfeng Hu: Writing – original draft, Methodology, Funding acquisition, Data curation, Conceptualization. **Sainan Wang:** Visualization, Validation, Investigation, Formal analysis. **Jiehui Zheng:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Qianwen Xu:** Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jretconser.2025.104264.

Data availability

Data will be made available on request.

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